

Vamsi Prasad

Full Stack Developer

• vamsi.p@mymailkeeper.com • +1(430)-236-1078 • LinkedIn • Open to Relocate

SUMMARY

Full Stack Developer with 4+ years of experience in designing, developing, and delivering web applications in the **finance** and **insurance domain** using **Java, Spring Boot, Hibernate, React.js, Node.js, Python, TypeScript, SQL, PostgreSQL, MongoDB**, and **RESTful APIs**, with **Kafka** and cloud deployment on **AWS**. Successfully developed mortgage and insurance platforms handling **30,000+** daily mortgage validations and **25,000+** weekly claims, improved backend query efficiency by **2.4×**, and reduced claim response time from **3.5 hours to 50 minutes**. Strong knowledge in large language models (LLMs) including **Transformers, BERT, GPT**, and **Generative AI** for intelligent application development.

SKILLS

Backend & API Technologies: Java 8/17, Spring Boot 3, J2EE, Spring MVC, Hibernate, JPA, JDBC, JSP, GraphQL, RESTful APIs, Apache Kafka
Frontend & UI Technologies: Angular 12+, React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive UI/UX Design
Cloud & DevOps: AWS (Lambda, EC2, S3, RDS, API Gateway), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Jenkins, Oracle Cloud
Databases & Optimization: PostgreSQL, MySQL, MongoDB, Redis, SQL Query Tuning & Performance Optimization
Tools & Version Control: IntelliJ IDEA, Git, GitHub, GitLab, Bitbucket, Swagger (API Documentation), Postman
Security & API Management: OAuth2, Spring Security, JWT, Role-Based Access Control (RBAC), API Gateway Management
Testing & Performance Validation: JUnit, Mockito, Selenium
Agile & Collaboration: JIRA, Confluence, Scrum/Kanban, Sprint Planning, Code Reviews
LLM & AI Technologies: Generative AI, BERT, GPT Transformers, LangChain

PROFESSIONAL EXPERIENCE

Freddie Mac, Dallas, Texas Jan 2024 – Present
Full Stack Developer

- Developed mortgage origination and loan-servicing modules using **Spring Boot, Hibernate**, and **React**, integrating underwriting, compliance, and borrower APIs, processing **30,000+** daily mortgage validations and improving traceability across lending workflows.
- Built event-driven microservices with **Spring Cloud** and **RabbitMQ** to handle credit, risk, and disbursement operations, improving inter-service communication throughput and reducing message queuing delays across financial systems.
- Implemented **ElasticSearch** and **Kibana** dashboards for real-time monitoring of borrower portfolios, enabling internal teams to search and visualize loan and repayment data within **milliseconds** across multiple regions.
- Enhanced data caching and persistence using **Hibernate** and **Redis**, reducing redundant read/write operations and improving backend query efficiency by **2.4×** for portfolio and payment history requests.
- Integrated authentication and access control through **Spring Security** and **OAuth2**, enforcing role-based authorization for internal staff, underwriters, and financial auditors in accordance with Freddie Mac’s compliance standards.
- Deployed microservices on **AWS ECS** using **Jenkins** and **Docker** pipelines, streamlining production rollouts and lowering deployment time per release to under **20 minutes**.
- Collaborated with data and DevOps teams to create centralized logging and monitoring through the **ELK Stack (Elasticsearch, Logstash, Kibana)**, establishing full traceability for loan-processing pipelines and compliance audits.
- Integrated Generative AI models using **LangChain** and **Hugging Face Transformers** to automate loan documentation summarization and risk report generation, reducing manual review time from **hours to minutes** and improving analyst productivity across departments.

Kpit Technologies, India May 2020 – Aug 2022
Full Stack Developer

- Led the design and development of a claims processing platform using **Spring Boot, Hibernate**, and **PostgreSQL**, enabling digital claim submissions across motor, health, and life insurance portfolios, handling **25,000+ claims weekly** with automated validation workflows.
- Created **RESTful APIs** in **Node.js** and **Spring Boot** to integrate insurer systems via CWS and Apana, reducing average claim response time from **3.5 hours to 50 minutes** through streamlined data synchronization between agents, customers, and assessors.
- Established monitoring using **Prometheus** and **Grafana**, creating real-time dashboards and alerting matrices for claims processing pipelines, API performance, and microservice health, reducing incident response time and improving system reliability.
- Engineered **Python** and **SQL scripts** with **GraphQL APIs** to consolidate claims, customer, and policy datasets, building internal matrices to support predictive analytics, anomaly detection, and operational reporting, enhancing decision-making for underwriting and claims teams.
- Architected dashboards for claims lifecycle tracking and settlement status using **Angular** and integrated with **CompiT Analytics** engine, supporting real-time monitoring of **60+ KPIs** used by underwriters and regional claim managers.
- Optimized database performance by refactoring **Hibernate** queries, indexing frequently accessed claim tables, and creating query optimization scripts, reducing execution time from **900ms to 180ms** during high-load claim submissions.
- Applied secure authentication and authorization using **OAuth2/JWT** across microservices, ensuring protected access to sensitive claimant data for **1,400+** active adjusters and partner agents.
- Partnered with QA and DevOps teams to automate regression and integration testing using **JUnit, Mocha**, and **Jenkins**, deploying services via **Docker** and **AWS ECS**, achieving consistent weekly deployments with zero downtime.
- Supported migration of legacy monolithic insurance modules to a distributed microservices architecture, ensuring smooth interoperability with third-party **TPA, CRM systems**, and external analytics tools.

EDUCATION

Master of Science in Data Science
University of North Texas, Denton, Texas

ACADEMIC PROJECTS

FinSecure Portal – Digital Banking API Platform

Tech Stack: Java, Spring Boot, Spring Security, JWT, MySQL, Docker, REST APIs

- Formed and developed a secure backend for digital banking operations, processing over **10,000 daily transactions** including onboarding, deposits, withdrawals, fund transfers, and real-time transaction tracking, ensuring reliable financial workflows.
- Carried out **JWT-based** authentication and **role-based authorization** with **Spring Security**, protecting **1,500+ customer and admin endpoints**, enforcing data security, and reducing unauthorized access incidents.
- Devised transactional Spring Data JPA repositories with service-level logic, maintaining consistency across **multi-step banking operations involving 5,000+ concurrent users**, preventing data conflicts and improving operational reliability.
- Containerized the platform using **Docker** and created automated **Postman** integration tests, reducing manual testing effort by **70%** and enabling faster, repeatable deployment across multiple environments.